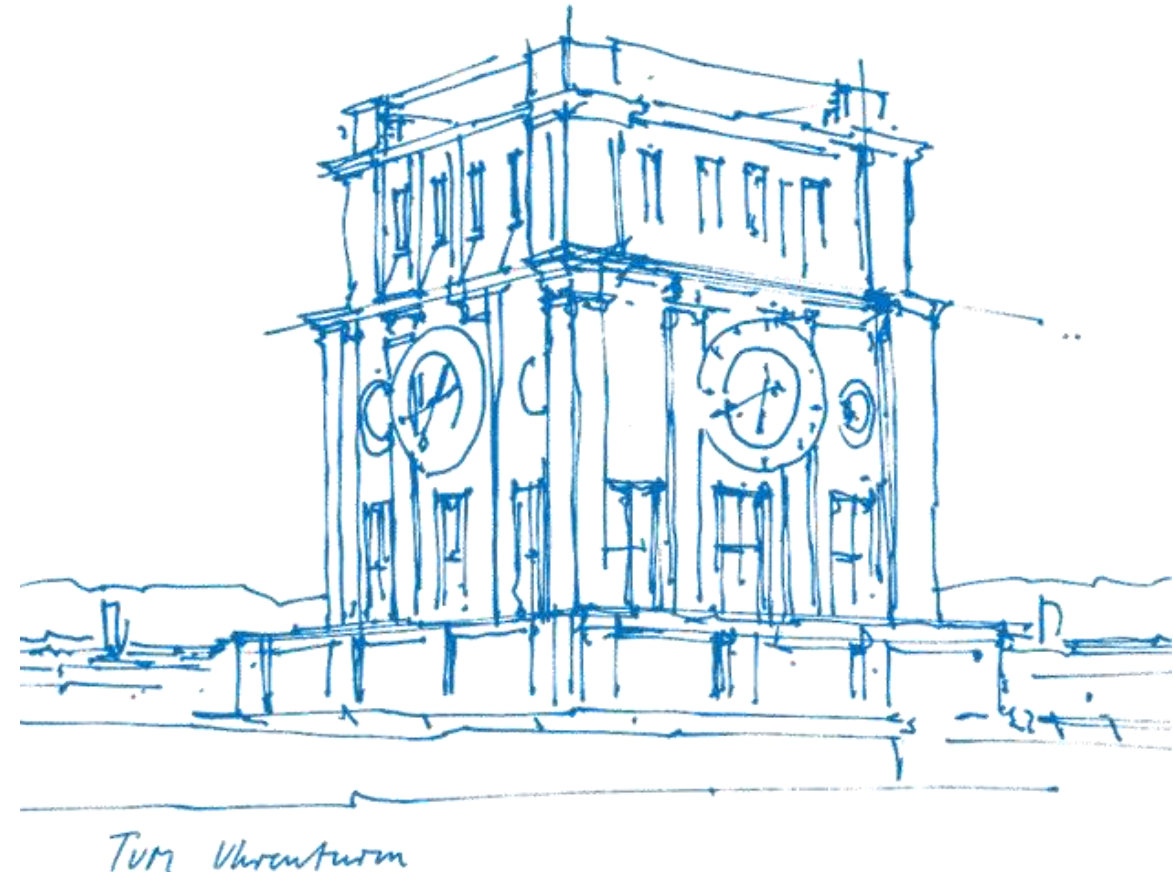


Tutorial 10

Transformation of Densities, Maximum Likelihood Estimation

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About Tutorial

■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

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- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- **Transformation of Densities**
 - Definition
- **Maximum Likelihood Estimation**
 - Definition
 - Unbiased Estimator

II. Exercise Sheet

Exercise 1 (sum of waiting times). The time (in minutes) a customer i has to wait for their coffee at a specialty coffee shop is modeled by an exponential random variable $X_i \sim \text{Exp}(\lambda)$ with $\lambda = 0.5$ (so the average wait is $1/\lambda = 2$ minutes). Let $S_{36} = \sum_{i=1}^{36} X_i$ be the total time required to serve 36 *independent* customers.

- (i) Find the characteristic function of a single waiting time X_i .
- (ii) Using the result from (i), what is the exact distribution of the total waiting time S_{36} ? State its name and parameters.
[Hint: Look at Example 2.50 of the lecture.]
- (iii) Use the clt to approximate the probability that the total waiting time for 36 customers exceeds 80 minutes.
[Hint: You will need the mean and variance of an $\text{Exp}(0.5)$ distribution. For the clt, you may use that for a standard normal RVRV Z , we have $P(Z > 2/3) \approx 0.2525$.]

II. Exercise Sheet

Exercise 2 (quality control in manufacturing). A factory produces resistors. The resistance of each resistor is intended to be 100 ohms. Due to manufacturing variability, the actual resistance values are iid random variables with mean $\mu = 100$ ohms and standard deviation $\sigma = 2$ ohms. A batch consists of $n = 64$ resistors. A batch is rejected if the average resistance of the 64 resistors is outside the range $[99.5, 100.5]$ ohms.

- (i) Let $\bar{X}_{64}$ be the average resistance of a batch of 64 resistors. What are the mean and variance of $\bar{X}_{64}$?
- (ii) Use Chebyshev's inequality to find an upper bound on the probability that a batch is rejected.
- (iii) Use the clt to approximate the probability that a batch is rejected.
[Hint: You may use that for a standard normal RVRV Z , we have $P(Z > 2) \approx 0.0228$.]

II. Exercise Sheet

Exercise 3 (transformation and convergence in distribution). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of RVRVs with $X_n \sim \text{Unif}(0, 1/n)$ for each $n \in \mathbb{N}_{>0}$. Let $Y_n = nX_n$.

(i) For a fixed n , find the pdf $p_{Y_n}(y)$ of Y_n .

[Hint: Use the change of variables formula.]

(ii) Find the cdf $F_{Y_n}(y)$ of Y_n .

(iii) Determine the limiting cdf $F_Y(y) = \lim_{n \rightarrow \infty} F_{Y_n}(y)$.

(iv) To what RVRV Y does Y_n converge in distribution? Specify its distribution.

II. Exercise Sheet

Exercise 4 (mle for uniform distribution). Let $X_1, \dots, X_n$ be iid RVRVs from a uniform distribution $\text{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i | \theta) = \frac{1}{\theta} \chi_{[0, \theta]}(x_i)$.

- (i) Write down the likelihood function $L(\theta | x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.
- (ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\text{mle}}$ for θ .
[Hint: The likelihood function may not be differentiable with respect to θ , but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]
- (iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\text{mle}}]$. Is $\hat{\theta}_{\text{mle}}$ an unbiased estimator for θ ?
[Hint: First find the cdf of $\hat{\theta}_{\text{mle}}$, then its pdf, and then compute the expectation.]

III. Additional Exercises

Exercise 5. The student Arnold wants to estimate the number n of fish in the lake behind the Faculty of Mechanical Engineering building. He devises the following method:

1. First, he catches a random sample of m fish, marks them appropriately, and releases them back into the lake.
2. After some time, he catches another random sample of k fish.
3. Let X be the number of marked fish in the second sample.

Objective: Determine a maximum likelihood estimator (MLE) for n based on X .

Hint: For optimizing the likelihood function, consider the ratio of two consecutive values of n , i.e.,

$$X \text{ is Hypergeometric-distributed, } P(X=x) = \frac{\binom{m}{x} \binom{n-m}{k-x}}{\binom{n}{k}}.$$

$$\text{With the given hint: } \frac{L(x; n)}{L(x; n-1)} = \frac{\binom{m}{x} \binom{n-m}{k-x}}{\binom{n}{k}} \cdot \frac{\binom{n-1}{k}}{\binom{m}{x} \binom{n-1-m}{k-x}} = \frac{(n-m)!}{(k-x)!(n-m-k+x)!} \cdot \frac{(n-1)!}{k!(n-1-k)!} \cdot \frac{k!(n-k)!}{n!} \cdot \frac{(k-x)!(n-1-m+k+x)!}{(n-1-m)!}$$

$$= \frac{(n-m)(n-k)}{n(n-m-k+x)} = \frac{n^2-nk-mn+nk}{n^2-nm-nk+nx} \leq 1, \quad n^2-nk-mn+nk \leq n^2-nm-nk+nx, \quad \hat{n} = \frac{mk}{x}.$$

Materials

